

Advancing Fashion Design Through Intelligent Sketchpad Studio

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Abstract

Line sketches serve as the visual DNA of fashion design, forming the essential foundation where concepts take shape, yet today's digital tools often lack the fluidity, personalization, and intelligence needed to truly support this creative process. We present FashSketch, an interactive, multimedia-driven system that reimagines fashion sketching through the lens of generative AI. Designed with a layer-based creative interface, FashSketch empowers designers to ideate, customize, and iterate on sketches seamlessly. By integrating state-of-the-art generative models, sketch-based retrieval, and large language models, the system supports advanced functionalities such as text-to-sketch generation and context-aware sketch recommendation. FashSketch not only enhances the sketching experience but also opens new multimodal pathways for creative expression, making it a powerful co-creative partner in the early stages of fashion design. Video demo is available at [here](#).

CCS Concepts

• **Information systems** → **Information retrieval**; • **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Media arts**; • **Computing methodologies** → **Computer vision**; **Machine learning**.

Keywords

Generative Model, Fashion Sketch, Fashion Retrieval, Visualization, Prompt Engineer, Fashion Generation

ACM Reference Format:

Nhu-Binh Nguyen-Truc, Nhu-Vinh Hoang, Tam V. Nguyen, Minh-Triet Tran, and Trung-Nghia Le. 2025. Advancing Fashion Design Through Intelligent Sketchpad Studio. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746027.3754481>

1 Introduction

Fashion sketching is the initial step in design, capturing early ideas and intended styles [20]. However, many lack the drawing skills to

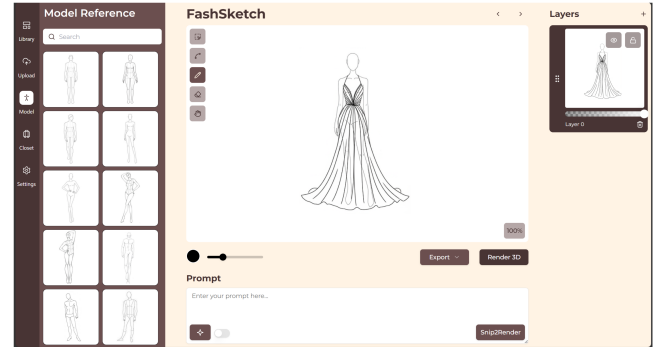


Figure 1: FashSketch's user interface.

translate concepts into sketches [4], creating a gap between vision and representation. This challenge is heightened in fashion, where designers must quickly generate multiple sketches, browse styles for inspiration, and conceptualize designs around a theme [5, 17]. A quick sketching tool would help preview shapes and speed up the design process. Early edge detection methods [3, 10] worked well on simple images but struggled with complex textures and lighting, often producing noisy results. CNNs [9] and neural style transfer [7] improved detail and artistic quality but were slow and sometimes missed fine details. More recently, diffusion-based models [15, 16, 21] have shown promise in sketch generation. Sketch-based systems [11, 12, 14] enable image generation from partial sketches with real-time refinement, though they are limited to incomplete sketches.

To tackle these challenges, we present FashSketch, an interactive multimedia system designed for both expert designers and novices. Featuring an intuitive, layer-based interface (Fig. 1), it facilitates precise editing and seamless customization, fostering creative exploration in fashion design. FashSketch integrates cutting-edge generative AI, sketch retrieval, and large language models to deliver advanced capabilities such as text-to-sketch search and dynamic sketch generation (Fig. 2). Leveraging a comprehensive dataset of 13,545 clothing images [19] augmented with additional web-crawled images and indexed via FAISS [6], the system provides a rich, searchable style library. Reference images are transformed into sketches using a HED edge detector [18], while a ControlNet-conditioned diffusion model [21] refines user sketches with context-aware guidance, enabling precise and adaptive design workflows. Video demo is available at [here](#).

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MM '25, Dublin, Ireland

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ACM ISBN 979-8-4007-2035-2/2025/10

<https://doi.org/10.1145/3746027.3754481>

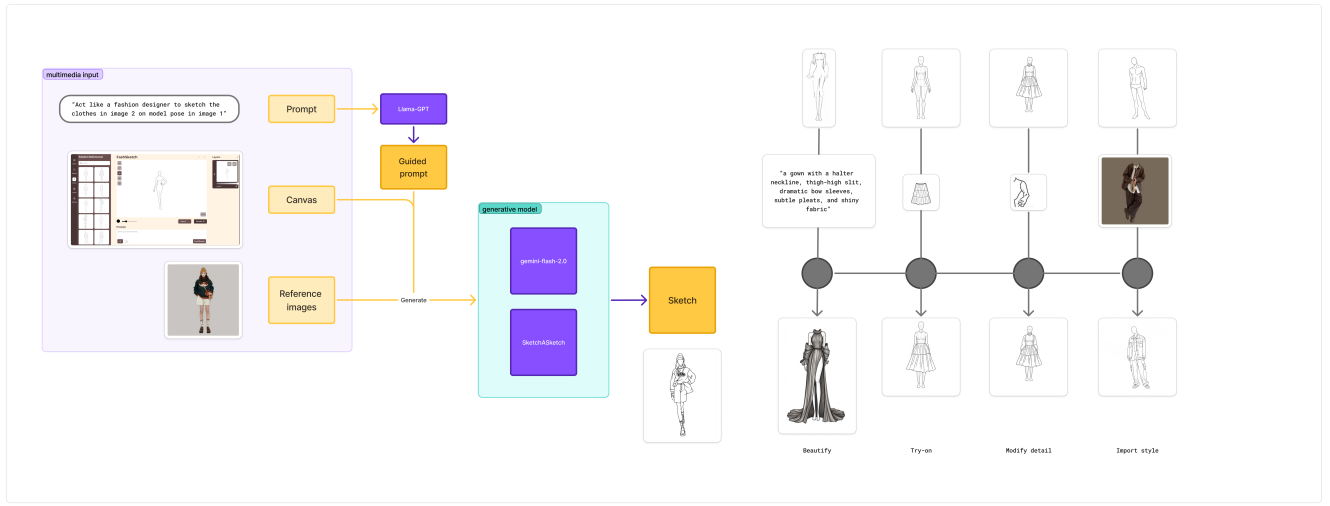


Figure 2: Workflow of the proposed FashSketch system

2 Proposed System

2.1 User Interface

We develop FashSketch, a system that supports early-stage design tasks such as finding inspiration, generating sketches, and refining concepts. FashSketch empowers users of all skill levels - from professionals to hobbyists - by offering intuitive tools that make fashion design more accessible and creatively engaging. The interface is designed to balance simplicity and functionality, making it approachable for beginners while retaining the flexibility professionals need. As shown in Fig. 1, it consists of four core components that streamline the design workflow.

The *Reference Sidebar* on the left offers searchable inspiration, including a default library, body models, closet samples, and image uploads that convert to sketches for consistent design foundations. At the center, the *Drawing Canvas* is the main workspace, featuring familiar edit tools like zoom, pencil, eraser, and selection to support intuitive sketching. Below it, export options allow downloads with transparent or white backgrounds. The *Prompt Input* at the bottom lets users enter natural language prompts and select canvas regions for AI-generated sketch suggestions, with built-in prompt refinement for better results. On the right, the *Layers Panel* enables locking, hiding, reordering, deleting, and adjusting layer transparency to manage complex designs efficiently.

2.2 Workflow

Figure 2 shows the FashSketch workflow, which unifies sketching, editing, visual search, and AI generation into a seamless process. The workflow begins on the canvas, where users can sketch directly, import existing files, or initiate design ideas using the system’s built-in visual search engine powered by CLIP (ViT/B32) and indexed with FAISS [6].

To refine a sketch, users select a canvas region and optionally add a prompt. This can be done to put reference clothes on a defined body pose sketch, elaborate the drawing, modify a detail of the garment. The system enhances input using Meta-Llama-3-8B-Instruct [1] and GPT-4 [8], then passes the prompt and region to

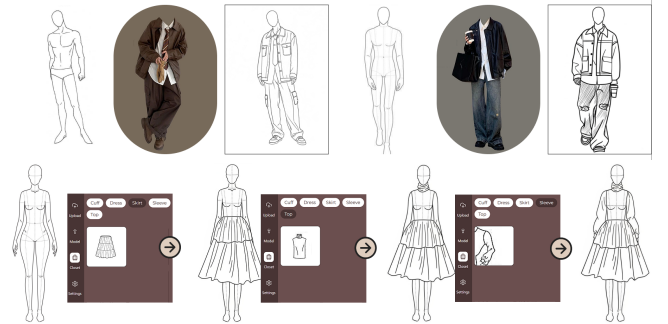


Figure 3: Our results using FashSketch for style importation, adding clothes and modify details

either a fine-tuned conditioned Stable Diffusion model [13, 14] or gemini-flash-2.0 [2]. This enables precise, intent-aligned sketch generation. Results are added as a new canvas layer, supporting iterative, non-destructive editing and a fluid, expressive creative workflow.

2.3 Qualitative Results

FashSketch supports concept visualization by preserving and enhancing user-drawn input through intelligent multimodal generation. The system interprets natural language prompts to produce adaptive base sketches that serve as starting points for further refinement (Fig. 3), reducing manual effort while maintaining creative control.

3 Conclusion

We presented FashSketch, an interactive system that enables autonomous generation of personalized fashion sketches through an intuitive layer-based interface. By integrating generative AI, sketch retrieval, and large language models, the system supports seamless multimodal interaction and efficient visual content creation. Future enhancements will focus on interface simplification, expanded sketching modalities, adaptive prompt guidance, and improved reference retrieval for more accurate and diverse outputs.

Acknowledgments

This research is supported by research funding from Faculty of Information Technology, University of Science, Vietnam National University - Ho Chi Minh City.

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